Apache Kafka — Core Concepts

1. Message

A **message** is a piece of data or an event that is sent to Kafka.

It represents information that you want Kafka to store and make available for processing.

Simple Definition

> A message is a record of data or an event sent to Kafka.

Example

```
"Ahmed placed an order for a laptop"
```

---

2. Producer

A **producer** is an application or system that sends messages to Kafka.

The producer decides which **topic** the message should be sent to.

Simple Definition

> A producer is an application that sends messages to Kafka topics.

Example

```
Order Application
        ↓
    Producer
        ↓
   orders topic
```

---

3. Topic

A **topic** is a named stream or category used to organize messages in Kafka.

You can think of a topic like a **folder for a particular type of data**.

For example:

```
orders
payments
customers
transactions
```

A topic is divided into **partitions**, and Kafka stores messages within those partitions.

Simple Definition

> A topic is a named stream or category used to organize and store messages in Kafka.

Example

```
Topic: orders

Message 1 → Ahmed placed an order
Message 2 → Sara placed an order
Message 3 → Ali placed an order
```

---

4. Consumer

A **consumer** is an application or system that reads messages from Kafka.

The consumer can then process the messages or send the data to another system.

Simple Definition

> A consumer is an application that reads and processes messages from Kafka topics.

Important

A consumer reading a message does **not normally remove the message from Kafka**.

Kafka keeps messages according to its retention configuration.

Example

```
orders topic
      ↓
  Consumer
      ↓
Process order
```

---

5. Broker

A **broker** is a Kafka server.

A Kafka cluster can contain multiple brokers, and each broker can store partitions from different topics.

Simple Definition

> A broker is a Kafka server that stores and serves messages.

Main responsibilities

A broker can:

- Receive messages from producers
- Store messages
- Serve messages to consumers
- Manage partitions
- Participate in replication

Example

```
Kafka Cluster

Broker 1
Broker 2
Broker 3
```

---

6. Partition

A **partition** is an ordered sequence of messages inside a Kafka topic.

A topic can contain one or many partitions.

Partitions allow Kafka to **scale and process data in parallel**.

Simple Definition

> A partition is an ordered sequence of messages within a topic that enables scalability and parallel processing.

Important

Messages within the **same partition are ordered**.

For example:

```
Partition 0

Message A
Message B
Message C
Message D
```

Kafka preserves this order:

```
A → B → C → D
```

However, you should not assume ordering across different partitions.

Example

```
Topic: orders

├── Partition 0
├── Partition 1
└── Partition 2
```

---

7. Offset

An **offset** is a sequential position number assigned to a message within a partition.

The offset identifies the position of a message in that partition.

Simple Definition

> An offset is the sequential position of a message within a Kafka partition.

Example

```
Partition 0

Offset 0 → Message A
Offset 1 → Message B
Offset 2 → Message C
Offset 3 → Message D
```

The offset allows a consumer to keep track of its position while reading messages.

For example:

```
Consumer processed:

Offset 0
Offset 1
Offset 2

Next message:

Offset 3
```

How the Concepts Work Together

```
                PRODUCER
                    |
                    | sends message
                    ▼
              ┌───────────┐
              │   TOPIC   │
              │  orders   │
              └───────────┘
                    |
          ┌─────────┼─────────┐
          ▼         ▼         ▼
     Partition  Partition  Partition
         0          1          2
         |          |          |
         └──────────┼──────────┘
                    |
              Kafka Brokers
                    |
                    ▼
                CONSUMER
```

Inside a Partition

```
Partition 0

Offset 0 → Message A
Offset 1 → Message B
Offset 2 → Message C
Offset 3 → Message D
```

Quick Definitions

```
Message:
A record of data or an event sent to Kafka.

Producer:
An application that sends messages to Kafka topics.

Topic:
A named stream or category used to organize messages.

Consumer:
An application that reads and processes messages from Kafka topics.

Broker:
A Kafka server that stores and serves messages.

Partition:
An ordered sequence of messages within a topic that enables
scalability and parallel processing.

Offset:
A sequential position number assigned to a message within a partition.
```

Mental Model

```
Producer
   ↓
Message
   ↓
Topic
   ↓
Partitions
   ↓
Broker
   ↓
Consumer
```

Remember

```
Producer → SENDS
Topic    → ORGANIZES
Partition → DIVIDES
Broker   → STORES / SERVES
```

```
Consumer → READS
Offset   → TRACKS POSITION
Message  → DATA / EVENT
```